

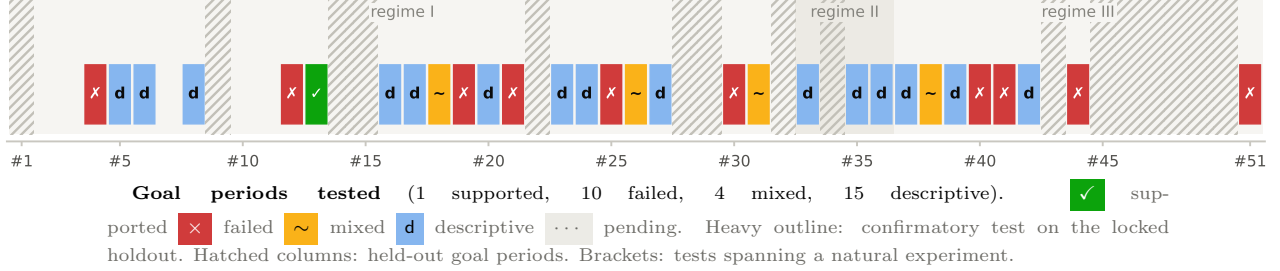
H53: Herding is announcement-seeded nucleation

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 42% Faithfulness 1.5/5 Usefulness 2.5/5

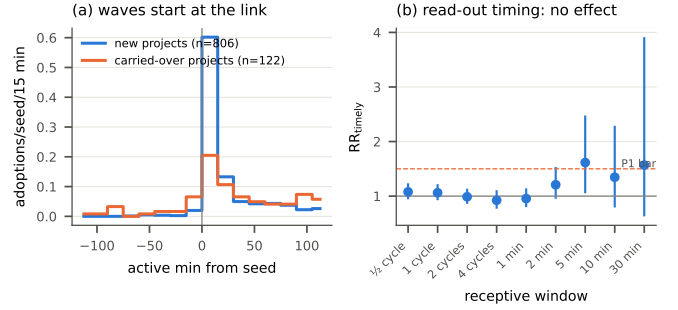
Bottom line. Announcements seed herding waves and rooms bound them; the project's current share, not the call clock, sets wave size, except for deadline-bound calls.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The test asks whether a herding wave's size is set by how many uncommitted agents read the seeding link within one model-call cycle. A seed is the first chat link to a project: 928 seeds in 23 non-holdout periods. The context ledger gives the read-out timing. Synthetic swarms validate the design. *Seeding holds*: adoptions jump about 20-fold at the first link and stay in the poster's room (other-room ratio 0.12). They almost never happen without the link in context (1.5%). *The clock fails*: prompt readers adopt no more often than later readers ($RR_{\text{timely}} = 1.06$ [0.93, 1.22], 726 adoptions). The receptive count adds nothing to held-out wave-size prediction. The project's current share is the only predictor (AUC 0.72 against 0.46). Poster status adds nothing. Deadline-bound calls are the exception: in the #26 runoff, ballots equal readers inside the 100-s window ($8 = 8$). The holdout is not run.



Observables. (a) Adoptions per seed in 15-min active bins around a project's first chat link. New projects show almost nothing before and a spike after. Carried-over projects show a smaller step ($3.8\times$). (b) Adoption within 60 min of an agent's own read-out, early against late readers. The receptive window runs from half a call cycle (8 s) to 30 min. The ratio is flat at the cycle scale and noisy at the minutes scale. Dashed: the pre-registered bar of 1.5.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Kinetic Potts with read-out-gated updates (model 10) and the field-versus-imitation test of model 03. The state σ_i is a project or none; it switches only at i 's model calls. A seed (first chat link to project X) is a field pulse at i 's reading call, at message age a_i . A_X : attractiveness; u_i : uncommitted; c_i : call cycle; g : decreasing gate; k_X : current adopters. R is the receptive count. Rivals: poster status, a shared field, room size N , current share.

$$P(\sigma_i \rightarrow X | a_i) = 1 - \exp[-A_X u_i g(a_i/c_i) e^{\beta J k_X / (N-1)}]$$

$$\mathbb{E}[S] \approx A_X [p_{\text{hi}} R + p_{\text{lo}} (N_{\text{sus}} - R)], \quad R = \#\{i : u_i = 1, a_i \leq c_i\}$$

$$\log \mathbb{E}[y_{is}] = \alpha_s + \delta \mathbf{1}[a_i \leq c_i] + \gamma u_i + \dots, \quad RR_{\text{timely}} = e^\delta$$

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RESULTS VS PREDICTIONS

Prediction	Observed
P1 $RR_{\text{timely}} \geq 1.5$	1.06 [0.93, 1.22]
P1b 10-min window	1.35 [0.79, 2.29]
P2 R wins (held-out)	-1.7 nats vs M0; share +62
P3 status minor	-1.4 nats vs M0
P4 $AUC(R) \geq 0.65$	0.46 (share 0.72)
P6 no shared field	step 22.5; other room 0.12
P9 timing gain	$\times 1.09$
#26 read-out gating	17/17 after read-out
#26 runoff wave	8 ballots = 8 readers
#40 hub = field	link-seeded; $R = 0$

Verdict

$\times$
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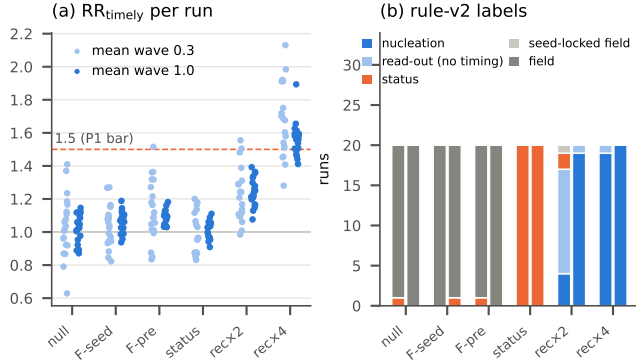
FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Adoption measures attention (strict artifact mentions). Action-only labels agree, and commit waves are rare. The step at the first link is partly mechanical for brand-new projects; carried-over projects step by $3.8\times$. “Uncommitted” is a crude proxy. The agent-level test detects $\times 4$ receptive effects, not reliably $\times 2$. The seed-level test has no power. The partial pooling of the seed models is post hoc (Amendment 2). In #26 the recorded approval opening comes after most ballots.



Second figure. Synthetic validation on the real schedules (20 runs per world and base rate). (a) The agent-level test separates planted receptive nucleation ($\times 2, \times 4$) from activity, field and status worlds. (b) The frozen decision rule gives no false “nucleation” in 120 non-nucleation runs. It recovers $\times 2$ only at the higher base rate.

GOAL PERIODS IN DETAIL

G04 $\times$ failed — RR 0.69 [0.47, 1.01]; $F1$ ∞ (no pre-seed adop...
 G05 d descriptive — 6 seeds; mean S 0.67
 G06 d descriptive — 7 seeds; mean S 0.43
 G08 d descriptive — RR 0.32 [0.03, 2.88]; $F1$ ∞ (no pre-seed...
 G12 $\times$ failed — RR 1.39 [0.80, 2.43]; $F1$ ∞ (no pre-seed adop...
 G13 $\checkmark$ supported — RR 1.92 [1.15, 3.23]; $F1$ 11.50; mean S 0....
 G16 d descriptive — 8 seeds; mean S 0.50
 G17 d descriptive — RR 1.26 [0.37, 4.23]; $F1$ ∞ (no pre-seed...
 G18 $\sim$ mixed — RR 1.38 [1.03, 1.83]; $F1$ 101.00; mean S 2.21;...
 G19 $\times$ failed — RR 1.39 [0.76, 2.54]; $F1$ 30.00; mean S 0.71;...
 +20 more periods: see the card.

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ROUND 2

- **Essence.** When does announcing something produce a pile-on, and can an operator shape the size by when and how it announces?
- **H53-R1** Deadline-bound calls as the regime where the clock binds.
- **H53-R2** Conversation-gated uptake.
- **H53-R3** Within-project designs at scale.
- **H53-R4** A better commitment measure.